Technical Report

Phishing Website Detection Using Machine Learning

COMP_3260 – Computer Network Security

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Abstract

Phishing websites are becoming a serious threat to internet users and phishing attacks via links in emails, messages, and advertisements are especially dangerous. This report presents the implementation of a machine learning (ML) model for phishing website detection based on features extracted from the HTML content of URLs. We used a dataset from Kaggle and trained a model using algorithms like logistic regression and decision trees to classify URLs as legitimate or phishing. Logistic regression achieved a high accuracy rate, which was offset by misclassification, and was corrected by the use of decision trees. The methodology, challenges, testing, future improvements such as browser extension for real time phishing detection are discussed in the report.

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1. Summary

In this report, we describe the development and implementation of a machine learning model to detect phishing websites based on HTML content analysis. A model is built which classifies URLs as legitimate or phishing by using a Kaggle dataset and extracting key features like HTTPS usage, presence of meta descriptions, number of images and JavaScript, social media links, and copyright information. Initially, the logistic regression model achieved 96% accuracy, but suffered from misclassification, which was resolved by changing to the decision tree model, resulting in 98% accuracy. Some challenges were the difficulty of accruing sufficient phishing URL data to test on and misclassifying certain URLs. Future work will include testing of other machine learning algorithms such as XGBoost, implementation of the model as a real time browser extension, and further testing to improve its accuracy and robustness.

2. Introduction.

Phishing websites are one of the main online security threats. By mimicking legitimate websites, they trick users into handing over sensitive personal information including passwords, credit card numbers, and other personal data. It is important to detect these phishing sites as phishing links are spread via email, social media, and other communication channels to protect internet users. This project aims at building a machine learning model for phishing website detection using the HTML content of URLs.

In this project the aim is to create a model that can correctly classify a website as legitimate or phishing according to different HTML features. In addition, the model would be experimented with different machine learning algorithms to make the accuracy of the model better and finally implement the model as a browser extension for real time phishing detection.

3. Literature Review.

The need for phishing detection has been increasing and many researchers have explored machine learning techniques to classify the phishing websites. Kathiravan et al. (2023) in his work titled *Detecting Phishing Websites using Machine Learning Algorithms*, gave a machine learning based approach of phishing detection. Kaggle dataset was used for the study and multiple algorithms such as decision Trees, random forest, k-nearest neighbors (KNN) and combinations of these algorithms were applied. In this study, features in the HTML structure of the URLs were identified and classification models were used to predict whether the website is phishing or legitimate. They deployed their model as a web service, where users could input URLs and the system would tell them whether or not it was phishing.

Similar to our project, we used the Kaggle dataset to train, and used decision trees and logistic regression for classification. Feature extraction and model training was based on the method from the reference study and we used different dataset, but we addressed issues such as misclassification and testing challenges, and improved the accuracy of the model.

Figure 1

The data set's first 15 rows

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	URL	Domain	TLD	URLSimilar	NoOfOther	SpacialCha	IsHTTPS	LineOfCode	Title	DomainTitl	URLTitleMa	IsResponsi	HasDescrip	HasSocialN	HasSubmit	HasCopyrig	NoOfImage	NoOfJS	NoOfSelfRe	label
2	https://ww	www.south	com	100	1	0.032	1	558	Ã Â¸â€šÃ Â	0	0	1	0	0	1	1	34	28	119	1
3	https://ww	www.uni-m	de	100	2	0.087	1	618	johannes gı	55.55556	55.55556	0	0	1	1	1	50	8	39	1
4	https://ww	www.voice	uk	100	2	0.069	1	467	voice fm so	46.66667	46.66667	1	1	0	1	1	10	7	42	1
5	https://ww	www.globa	org	100	1	0.033	1	1210	gri - home	0	0	1	1	1	0	1	35	11	86	1
6	https://ww	www.nerds	com	100	1	0.04	1	514	nerds cand	100	100	1	1	1	1	1	24	22	36	1
7	https://ww	www.hyder	in	100	1	0.034	1	2371	hyderabad	100	100	1	1	1	1	1	71	9	40	1
8	https://ww	www.aap.o	org	100	1	0.056	1	2730	home	0	0	1	1	1	0	1	10	12	173	1
9	https://ww	www.religic	com	100	1	0.03	1	2616	religiÃƒÂ³n	55.55556	55.55556	1	1	1	1	1	80	10	376	1
10	http://ww	www.teram	com	82.64463	1	0.045	0	2	0	0	0	0	0	0	0	0	0	0	0	0
11	https://ww	www.aoh6	com	100	1	0.05	1	5966	0	0	0	1	0	1	0	0	16	7	4	1
12	https://ww	www.bulga	com	100	1	0.038	1	2639	bulgaria ski	100	100	1	0	1	1	1	63	22	156	1
13	https://ww	www.bright	com	100	1	0.042	1	5509	0	0	0	1	0	0	1	1	12	19	2	1
14	https://ww	www.motle	ie	100	1	0.05	1	2839	home | mo	100	100	1	0	1	1	1	56	16	235	1
15	https://ww	www.funzir	hu	100	1	0.048	1	1576	funzine	100	100	1	1	1	1	1	52	27	0	1

4. Methodology

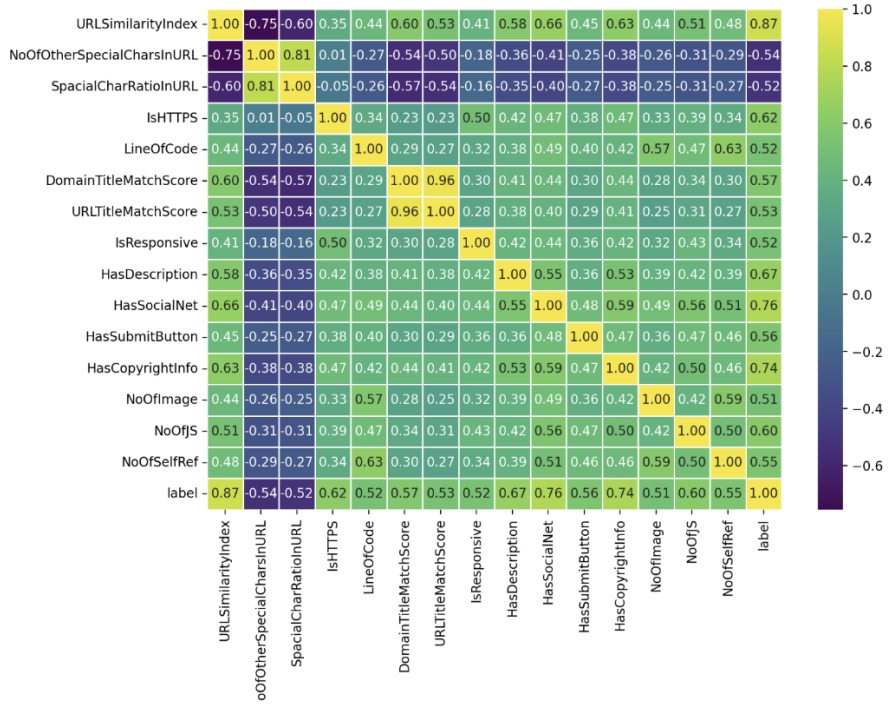
The development of the machine learning model for phishing detection was done in a systematic way. Me and my colleague worked on a different parts of the model, I worked on feature extraction and implementation of logistic regression while my colleague worked with dataset, visualization and implementation of decision trees model.

4.1 Feature Extraction

Key features taken were based on the weight of the feature on the label. The key features used in our model are ones of the features which have the most of the weight value effect on the "label" value(see Figure 2) that determines whether URL is legit or not.

Figure 2

The heat map of the data set

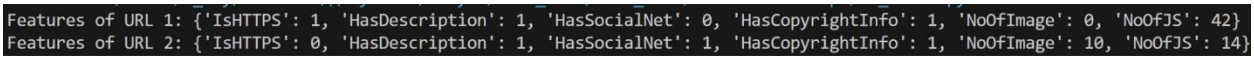


The analyze_html function implemented the key feature extraction process. It takes an HTML content of a given URL and parses it to extract features that may be able to identify phishing websites. The following steps were performed:

- HTML Parsing: Using BeautifulSoup library the HTML content of the website was parsed. Then, I used urlparse function from urllib.parse module to check if URL was HTTPS.
- Feature Identification:
    - HTTPS: If the website doesn't use HTTPS, which is often a red flag for phishing sites, the function flags it.
    - Description: We check for the presence of a description tag because phishing sites often don't have it or don't configure it properly.
    - Images and JavaScript: The tags <img> and <script> are counted. The more images and JavaScript tags the higher the number, the more complex or suspicious website it might be.
    - Social Media Links: This checks for the presence of links to popular social media platforms such as Facebook, Twitter, Instagram, and LinkedIn as phishing websites often try to make their site mimic a legitimate site.
    - Copyright Information: Legitimate websites often include copyright information in the HTML, so the function searches for it.

Figure 3

The calculated values of HTML



4.2 Machine Learning Models

As this is a common algorithm used for binary classification tasks, we first tested the logistic regression model on the extracted features. But the results were suboptimal. The model attained a 96% accuracy(see Figure 3), however, it misclassified several URLs. In particular, when both a valid URL and a phishing URL were input, the model misclassified both as valid. When I used the predict_proba function to get the probability of phishing, the result for a phishing URL was just 1%, which means that the model wasn't very confident in its prediction.

Figure 4

The results of the logistic regression model

```
Logistic Regression Predictions: [1 1 0 1 1 0 0 0 0 1]
Logistic Regression Accuracy: 0.9634902804098
```

To overcome this problem, we moved to decision trees, which greatly increased the classification accuracy to 98%(see Figure 4). Decision trees are well known in their ability to deal with non linear relationships between features and are often more interpretable than other models. Results from the Decision Tree model were much better, obtaining higher accuracy in the identification of phishing and legitimate websites.

Figure 4

The results of the decision trees model

```
Accuracy: 0.9841148271069129
Confusion Matrix: [[20539   306]
 [  254 14154]]
```

5. Results and Discussion

The logistic regression and decision tree models were trained on the dataset and performance was evaluated using accuracy as the main metric. The logistic regression model gave an accuracy of 96% but URLs were being misclassified. Especially when the same website was input as a phishing website and a legitimate website, both were marked as legitimate.

We used the predict_proba function to see the scores of probability of each URL so we could diagnose the problem. For the phishing website, the probability of being phishing was predicted to be 1% which was clearly wrong. It implied that the Logistic Regression model could not be very sensitive to find out phishing websites.

Figure 5

The input URLs

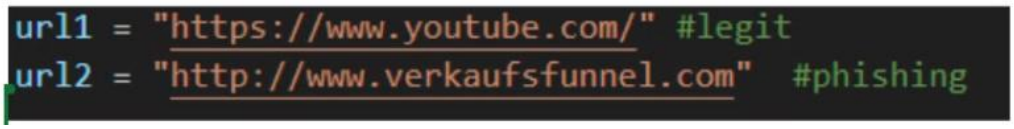
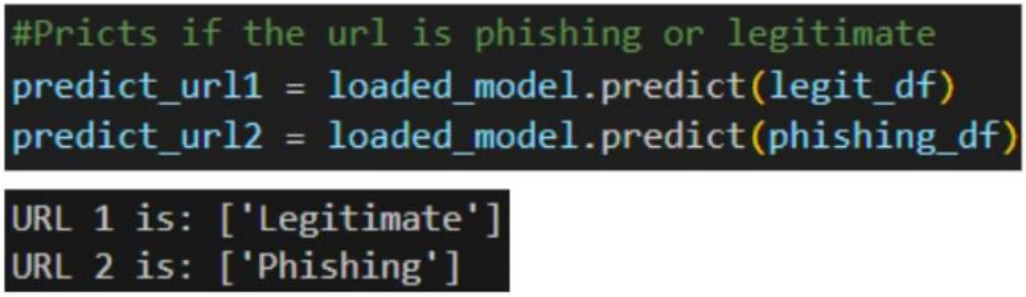


Figure 6

The output of the model



On the other hand, the decision tree model performed much better with an accuracy of 98%. It correctly classified phishing websites and legitimate websites(see Figure 5 and 6), and it was

better suited for this task. The higher performance of the decision tree confirmed that the non linear models could have better performance on this classification problem.

6. Conclusions and Future Work

6.1 Accomplishments

A machine learning model of phishing website detection has been realized in this project. The model analyzed the HTML content of URLs and extracted key features, to classify URLs as legitimate or phishing. The result shows that the decision tree model has better accuracy and classification performance compared to the logistic regression model.

6.2 Challenges

A major challenge during the project was the misclassification of URLs using logistic regression. While the model was able to achieve high accuracy, it was not able to correctly identify phishing websites in some test cases. Furthermore, it was difficult to get a sufficiently large dataset of phishing URLs to test from.

6.3 Future Work

Other machine learning algorithms, including XGBoost and random forest, will be explored to further improve the model and potentially have a better performance on detecting the phishing websites. Further, the next step will be to implement the model as a real time browser extension. This extension will send URLs to a server where the model will classify the site and if the URL is detected as phishing a pop up notification will be displayed. Finally, the model needs more testing to refine it and make sure it is robust.

References

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Prishasawhney. (2023). Phishing URL Website Dataset (Cleaned). *Kaggle*. Retrieved from https://www.kaggle.com/datasets/prishasawhney/phishing-url-website-dataset-cleaned/data

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